

Vascular Resistance Network Analysis

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1.1 Skeleton-to-Centerline Graph Construction

A dense one-dimensional (1-D) centerline graph was constructed directly from the skeletonized vessel mask. Every non-zero voxel in the binary skeleton volume was assigned a unique node positioned at its world-coordinate voxel centre, derived from the NIfTI affine transformation $\mathbf{A}$:

$$\mathbf{x}_k = \mathbf{A} \begin{bmatrix} i_k \\ j_k \\ k_k \\ 1 \end{bmatrix}_{[1:3]}, \quad k = 1, \dots, N, \quad (1)$$

where (i_k, j_k, k_k) are the zero-indexed voxel coordinates of node k and N is the total number of skeleton voxels. Undirected edges were placed between every pair of 26-connected adjacent skeleton voxels, so the mesh topology exactly mirrors the skeleton connectivity in the original volume.

1.2 Vessel Radius Estimation

The local vessel radius at each node was estimated as the Euclidean distance from that node to the nearest background voxel. A distance transform was computed on the binary vessel mask Ω using the method of Meijster et al. [1]:

$$r(\mathbf{x}) = \text{EDT}(\mathbf{1} - \mathbf{1}_\Omega)(\mathbf{x}), \quad (2)$$

where the transform was evaluated with per-axis voxel spacing $(\Delta x, \Delta y, \Delta z)$ to correctly handle anisotropic acquisitions. The resulting distance-transform volume was then sampled at each node world position via nearest-voxel interpolation (using the inverse affine), and values below 0.1 mm were clamped to that floor to avoid degenerate conductances.

1.3 Hagen-Poiseuille Conductance

Each edge $e = (i, j)$ was assigned a lumped Hagen-Poiseuille conductance

$$G_{ij} = \frac{\pi \bar{r}_{ij}^4}{8 \mu L_{ij}}, \quad (3)$$

where $\bar{r}_{ij} = (r_i + r_j)/2$ is the mean nodal radius along the edge, $L_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|_2$ is the Euclidean edge length, and $\mu = 3.5 \times 10^{-3} \text{ Pa} \cdot \text{s}$ is the dynamic viscosity of blood. All quantities were converted to SI units before evaluation.

1.4 Phantom Gap Edges

Skeletonization and image acquisition artifacts frequently produce multiple disconnected components. To obtain a fully connected conductance network, phantom gap edges were inserted between components separated by less than a user-specified maximum gap distance $d_{\max}$.

Candidate endpoint selection. For each connected component, candidate gap-bridging nodes were identified as degree- ≤ 1 skeleton tips (leaves); for isolated single-node components all nodes were considered candidates.

MST bridging (default). A k -d tree was built on all candidate nodes. All cross-component candidate pairs within Euclidean distance $d_{\max}$ were enumerated via `query_pairs`. For each pair of components (c_i, c_j) , only the closest candidate pair was retained as the representative bridge, giving a component-level weighted graph. A minimum spanning tree (MST) was computed on this component graph to select the minimal set of bridges that ensures full connectivity. This guarantees exactly one phantom edge per gap at minimum total phantom length, avoiding spurious parallel conductance paths that would distort the flow solution.

k-NN bridging (fallback). As an alternative, every cross-component candidate pair within $d_{\max}$ received a phantom edge. This matches the behaviour of the legacy MATLAB implementation but can introduce redundant parallel paths when many tip nodes are spatially clustered.

Phantom conductance. Each phantom edge (i, j) was assigned a penalised conductance

$$G_{ij}^{\text{ph}} = \frac{\pi \bar{r}_{ij}^4}{8 \mu \alpha L_{ij}}, \quad (4)$$

where $\alpha = 10$ is a dimensionless penalty factor ensuring that phantom bridges carry negligible flow relative to real vascular connections.

1.5 Boundary Conditions

Boundary conditions were assigned automatically from the real-graph topology. The inlet node was defined as the degree- ≤ 1 leaf with the largest local radius (reflecting the largest visible feeding vessel), and prescribed a pressure $p_{\text{in}} = 100$ mmHg. All other real-graph leaves were treated as outlets and prescribed $p_{\text{out}} = 5$ mmHg. In the absence of any topological leaves (e.g. fully cyclic sub-graphs), the nodes with the maximum and minimum z -coordinates were used as inlet and outlet, respectively.

1.6 Sparse Conductance System and Solution

The nodal pressure field $\mathbf{p} \in \mathbb{R}^N$ was obtained by solving the symmetric sparse Kirchhoff system

$$\mathbf{K} \mathbf{p} = \mathbf{q}, \quad (5)$$

where the global conductance (Laplacian) matrix is

$$K_{kl} = \begin{cases} \sum_{j \sim k} G_{kj} & k = l, \\ -G_{kl} & k \neq l, (k, l) \in \mathcal{E}, \\ 0 & \text{otherwise,} \end{cases} \quad (6)$$

and $\mathcal{E}$ is the union of real and phantom edges.

Boundary conditions were incorporated via Schur-complement elimination. Let $\mathcal{F}$ and $\mathcal{C}$ denote the sets of free and constrained nodes, respectively. The block system

$$\begin{bmatrix} \mathbf{K}_{FF} & \mathbf{K}_{FC} \\ \mathbf{K}_{CF} & \mathbf{K}_{CC} \end{bmatrix} \begin{bmatrix} \mathbf{p}_F \\ \mathbf{p}_C \end{bmatrix} = \mathbf{0} \quad (7)$$

was reduced to

$$\mathbf{K}_{FF} \mathbf{p}_F = -\mathbf{K}_{FC} \mathbf{p}_C, \quad (8)$$

which was solved with a direct sparse solver. Any connected component of the combined graph that contained no boundary node was pinned to the mean pressure $p_{\text{mean}} = (p_{\text{in}} + p_{\text{out}})/2$ to prevent a singular system. Zero-diagonal rows arising from fully isolated nodes after partitioning were regularised by setting the diagonal to unity and the right-hand side to p_{mean} .

1.7 Volumetric Flow

The volumetric flow through each real edge was computed from the pressure solution as

$$Q_{ij} = G_{ij} (p_i - p_j), \quad (9)$$

with results reported in mm^3s^{-1} . The full pressure, radius, flow, conductance, and edge-length fields were written back to the centerline mesh in VTK PolyData format (VTP) for visualisation in ParaView.

References

- [1] Arnold Meijster, Jos B.T.M. Roerdink, and Wim H. Hesselink. A general algorithm for computing distance transforms in linear time. *Mathematical Morphology and its Applications to Image and Signal Processing*, pages 331–340, 2000.